

Victorian Biodiversity Research Portfolio

Biodiversity & Ecosystems Unit

Year 7 | Victorian Curriculum F-10 v2.0

11 Stages

How this portfolio prepares you for the CAT

The Common Assessment Task has seven parts, **A** to **G**. Every stage below feeds directly into one or more of them. If you find a CAT question hard, come back to the stage listed beside it.

In the CAT you get a **Species Booklet** for your species. In this portfolio you get a **Species Profile** for the Eastern Barred Bandicoot, laid out in the same five sections — *Profile, Ecology, Food-web evidence, Threats, Conservation*. Learning where to find things in it is half the practice.

Stage	Portfolio topic	Prepares you for
1	Living or Non-living? (MRS GREN)	A1 living / dead / non-living • B2 respiration
2	Classification	A2 why we classify • A4 classification table
3	Scientific Names & Six Kingdoms	A3 two-part naming and its rules
4	Dichotomous Keys	A5 identifying Specimen X and justifying it
5	Australian Ecosystems & Biodiversity	D1–D3 biotic & abiotic factors • E1 why biodiversity matters
6	Food Webs & Energy Flow	B1–B4 photosynthesis, respiration, producer/consumer • C1–C5 the whole food-web part
7	Ecosystem Services	E1 value of biodiversity • E4 effect of losing a species
8	Adaptations	D3 how factors affect a species • G poster content
9	Introduced Species	E2 defining “invasive” • E3 impact on the species
10	Human Impacts on Ecosystems	D4 predicting flow-on effects of a change
11	Conservation & Action	F1–F3 weighing strategies and justifying a recommendation

The part that matters most

Part **F** — *Conservation: Evaluate & Recommend* — is the key criterion of the CAT. It asks you to weigh three strategies against each other, recommend one, justify it with evidence, and admit one weakness of your own choice. Stage 11 and Stage 12 both rehearse exactly this move.

Our Class Species

Eastern Barred Bandicoot

Perameles gunnii

A grassland digger that came back from extinction in the wild

Why the whole class studies the same species

Everyone in this class investigates the **Eastern Barred Bandicoot**. That means we can compare answers, argue about the same food web, and disagree about the same conservation strategies — which is exactly what makes the thinking sharper.

Your facts come from the **Species Profile** booklet, which is laid out in the same five sections as the Species Booklet you will be given in the CAT: *Profile*, *Ecology*, *Food-web evidence*, *Threats* and *Conservation*. Keep it beside you.

The CAT uses a **different species**. You will not be handed the bandicoot on the day. What transfers is the *method* — how to read a species booklet, build a web from a list of organisms, and weigh three strategies against a threat.

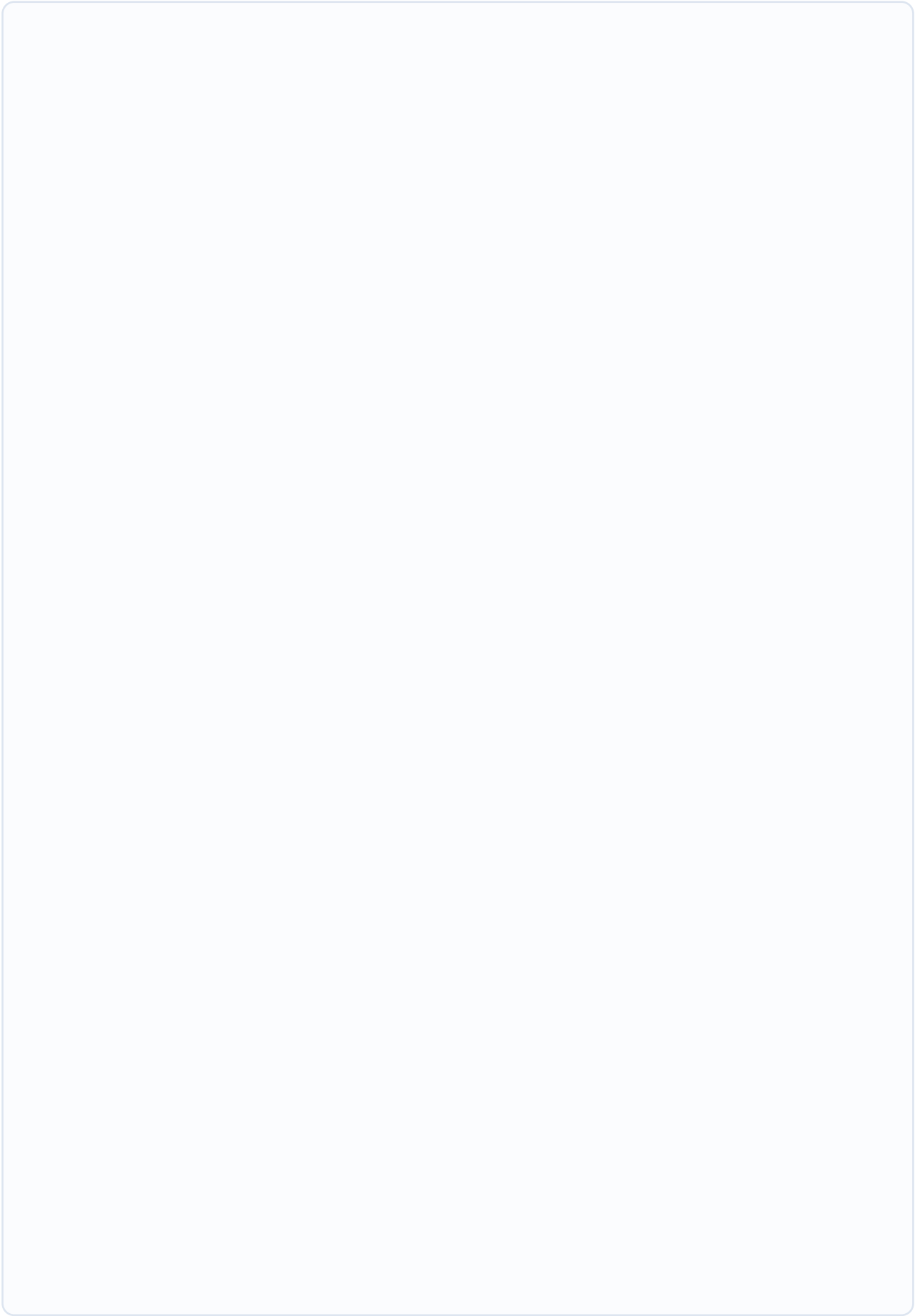
Common name	Eastern Barred Bandicoot (mainland)
Scientific name	<i>Perameles gunnii</i>
Ecosystem	Native grassland and grassy woodland
Victorian region	Volcanic plains of western Victoria; islands in Western Port
Conservation status	Endangered — reclassified from <i>Extinct in the Wild</i> in 2021

— Before we start

Fill this in from the Species Profile

Write **three** things about this animal that surprised you, and **one** question you want answered by the end of the unit.

Attempt a scientific sketch of the Eastern Barred Bandicoot:



1

Stage 1: Living or Non-living?
Lesson concept: MRS GREN — the seven life processes

Background

All living things carry out the seven life processes summarised by the acronym **MRS GREN**: Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, and Nutrition. In this stage you will demonstrate, with specific evidence, that your chosen organism is alive.

— At The Level

Released: Lesson 1

Where to look: Unit Booklet, Section 1 — Living or Non-living? • **Species Profile: Ecology**

Complete the table below for the Eastern Barred Bandicoot. In the **Evidence** column, give a *specific* example — not just the name of the process (e.g. not “moves” but “digs hundreds of small conical pits in a single night to reach worms in the soil”).

Life process	Specific evidence from the bandicoot
Movement	
Respiration	
Sensitivity	
Growth	
Reproduction	
Excretion	
Nutrition	

— Above The Level

Released: Lesson 1

Identify ONE life process that is most difficult to observe directly in the wild for the bandicoot. Explain why it is hard to observe, and describe how scientists might use technology or indirect evidence to detect it.

2

Stage 2: Classification

Lesson concept: The Linnaean system — seven levels of taxonomy

Background

Scientists classify all living things into a hierarchy of groups from broadest to most specific: Domain, Kingdom, Phylum, Class, Order, Family, Genus, Species. In this stage you will place the bandicoot within that system and explain what each level tells us about its evolutionary relationships.

— At The Level Released: Lesson 2

Where to look: Unit Booklet, Section 2 — Linnaean Levels • **Species Profile:** Profile (Kingdom to Species) and **Names & Classification** (Domain)

Part A — Complete the Linnaean classification table for the bandicoot.

Level	Name for the bandicoot	What this level means (in your own words)
Domain		
Kingdom		
Phylum		
Class		
Order		
Family		
Genus		
Species		

Where to look: Species Profile: Related Species

Part B — Name ONE other organism that shares the same *Order* as the bandicoot. State one physical or behavioural feature they share that reflects their common ancestry.

— Above The Level

Released: Lesson 2

Where to look: Species Profile: Related Species

Compare the bandicoot to another species in the same Order but a different Family — the Species Profile names one. Identify TWO differences between them and explain what those differences tell us about how the two lineages have diverged over time.

Where I found this — name the Species Profile section or Unit Booklet page for each answer above.

3

Stage 3: Scientific Names & The Six Kingdoms

Lesson concept: Binomial nomenclature and kingdom-level features

Background

Every species has a two-part Latin scientific name (binomial nomenclature) consisting of its Genus and species epithet. This universal system prevents confusion when scientists across the world discuss the same organism. All life is also grouped into one of six kingdoms, each with distinctive cellular and structural characteristics.

— At The Level

Released: Lesson 3

Where to look: Species Profile: Names & Classification

Part A — Find the meaning and origin of the bandicoot’s scientific name.

Component	Meaning / origin
Genus name	
Species epithet	
Why this name was chosen	

Where to look: Unit Booklet, Section 3 — The Six Kingdoms • Species Profile: Profile

Part B — State which of the six kingdoms the bandicoot belongs to. Give TWO kingdom-level characteristics that apply to it.

— Above The Level

Released: Lesson 3

Where to look: Species Profile: Names & Classification — the “When scientific names change” box

Scientific names are sometimes revised after new evidence. Using the example in your Species Profile, explain what evidence caused the change, and why updating names matters — not just for scientists, but for whether a species gets protected at all.

4

Stage 4: Dichotomous Keys

Lesson concept: Identifying organisms using observable features

Background

A dichotomous key guides users through a series of either/or questions based on observable physical features, allowing them to identify an unknown organism. Keys must use precise, observable characteristics – not size, colour, or behaviour unless these are constant across the species.

— At The Level Released: Lesson 4

Where to look: Unit Booklet, Section 4 — Dichotomous Keys • **Species Profile: Related Species** — use the three other animals listed there and their observable features

Design a **table-format** dichotomous key that distinguishes the bandicoot from the THREE other animals on the Related Species page.

The four organisms in my key:

1. _____ (Eastern Barred Bandicoot)
2. _____
3. _____
4. _____

Complete the key below. Each step must lead either to an organism name (end) or to another step number. Use only *observable structural features*.

Step	Feature / question	Go to / Result
1a		
1b		
2a		
2b		
3a		
3b		
4a		
4b		

Now construct a tree for the above dichotomous key.

— **Above The Level**

Released: Lesson 4

Identify ONE limitation of your dichotomous key. Describe a specific situation — what type of specimen, life stage, or condition — that would cause the key to give a wrong or impossible result. Suggest how the key could be improved to address this limitation.

5

Stage 5: Australian Ecosystems & Biodiversity

Lesson concept: Biotic and abiotic factors; species richness

Background

An ecosystem is a community of living organisms (biotic factors) interacting with each other and with the non-living environment (abiotic factors) in a specific area. Australia’s ecosystems are among the world’s most biodiverse, shaped by millions of years of geographic isolation.

At The Level

Released: Lesson 5

Where to look: Unit Booklet, Section 5 — Australian Ecosystems • **Species Profile:** Ecology, Food-web evidence and Threats

Part A — Identify the ecosystem the bandicoot lives in and list key biotic and abiotic factors.

Ecosystem type: _____

Biotic factors (at least 4)	Abiotic factors (at least 4)

Part B — Choose ONE abiotic factor and explain specifically how it shapes which organisms can survive in this ecosystem.

— Above The Level

Released: Lesson 5

Where to look: Unit Booklet, Section 5 — Australian Ecosystems (choose a second ecosystem described there) • **Species Profile: The Grassland**

Compare the biodiversity of the bandicoot's grassland with ONE other Victorian ecosystem described in your Unit Booklet. Use *species richness* as your primary measure. Explain what *abiotic* differences between the two account for the difference — and say whether the ecosystem with fewer species is therefore less worth protecting.

6

Stage 6: Food Webs & Energy Flow*Lesson concept: Trophic levels and energy transfer***Background**

Energy enters most ecosystems through photosynthesis and flows from producers to consumers through feeding relationships. A food web shows the interconnected feeding relationships in an ecosystem. Only approximately 10% of energy transfers between each trophic level.

— At The Level

Released: Lesson 6

Where to look: Unit Booklet, Section 6 — Food Webs • **Species Profile:** Food-web evidence — the organism list is there; you work out the arrows

Part A — Draw a food web for the bandicoot's grassland below. Include at least **8 organisms** with the bandicoot clearly labelled. Use arrows to show energy flow (arrow points from food source to consumer).

Part B — Identify which level consumer the Bandicoot is in your food web.

— **Above The Level**

Released: Lesson 6

Part A — Trace a complete energy pathway from a *producer* to the bandicoot, naming each organism and trophic level.

Part B — Predict what would happen to TWO other species in your food web if the bandicoot were completely removed. Use the concept of a *trophic cascade* to explain each effect. Specify whether the other species would increase or decrease in number and why.

7

Stage 7: Ecosystem Services

Lesson concept: What ecosystems and species provide for people

Background

Ecosystem services are the direct and indirect benefits that ecosystems and the organisms within them provide to humans. They are grouped into four categories: **provisioning** (food, water, materials), **regulating** (climate, water purification, pollination), **cultural** (spiritual, recreational, educational), and **supporting** (soil formation, nutrient cycling, primary production).

At The Level

Released: Lesson 7

Where to look: Unit Booklet, Section 7 — Ecosystem Services • **Species Profile:** The Grassland and Ecology

Identify THREE ecosystem services provided by the bandicoot or its grassland. For each service: name it, classify it (provisioning / regulating / cultural / supporting), and give specific evidence of how it benefits people or other species.

#	Service name	Category	Specific evidence / explanation
1			
2			
3			

Above The Level + First Nations Perspectives

— Perspectives

Released: Lesson 7

Where to look: Species Profile: First Nations Knowledge

Using the Species Profile, describe ONE specific practice or relationship through which First Nations peoples have cared for these grasslands. Classify it as an ecosystem service, and explain how this knowledge informs land management in Victoria today.

8

Stage 8: Adaptations

Lesson concept: Structural, behavioural, and physiological adaptations

Background

An adaptation is a feature that improves an organism’s survival or reproductive success in its environment. Adaptations arise through natural selection over many generations. They are classified as **structural** (body form), **behavioural** (actions), or **physiological** (internal processes).

— At The Level Released: Lesson 8

Where to look: Unit Booklet, Section 8 — Adaptations • **Species Profile: Ecology** — five adaptations are listed; you decide which type each one is

Choose THREE adaptations of the bandicoot — one of each type. For each: name it, describe it precisely, and explain specifically how it increases survival or reproductive success in the bandicoot’s environment.

Type	Name & description	How it increases survival / reproduction
Structural		
Behavioural		
Physiological		

— Above The Level

Released: Lesson 8

Choose ONE of the adaptations above. Explain, step by step, how natural selection would have led to this adaptation over many generations. Your answer must include: variation in the original population, the selection pressure, differential survival, and change over generations.

9

Stage 9: Introduced Species

Lesson concept: Threats to native biodiversity from invasive species

Background

Introduced (invasive) species are organisms moved outside their natural range, often by human activity. In Australia, introduced species such as foxes, cats, rabbits, and cane toads have devastated native wildlife. They compete for resources, prey on native species, spread disease, and alter habitats.

— At The Level

Released: Lesson 9

Where to look: Unit Booklet, Section 9 — Introduced Species • **Species Profile: Threats, Timeline & Status and Conservation**

Choose ONE introduced species that threatens the bandicoot or its ecosystem.

Introduced species:

Question	Your answer
How was this species introduced to Australia? (pathway, date/era, reason)	
What specific ecological impact does it have on the bandicoot or native species in the ecosystem?	
What is ONE control method currently being used? Describe how it works.	

— Above The Level

Released: Lesson 9

Evaluate the control method you described above. What evidence shows it is reducing the impact of the introduced species? What are its key limitations (ecological, economic, or practical)? Is there a better alternative? Use specific evidence.

Where I found this — name the Species Profile section or Unit Booklet page for each answer above.

10

Stage 10: Human Impacts on Ecosystems

Lesson concept: *Habitat loss, pollution, climate change, and over-exploitation*

Background

Human activities affect biodiversity through habitat destruction and fragmentation, pollution, over-exploitation of species, and climate change. These impacts reduce species richness, disrupt ecosystem services, and can push species toward extinction. Understanding the mechanisms of impact is the first step to reducing them.

— At The Level

Released: Lesson 10

Where to look: Unit Booklet, Section 10 — Human Impacts • **Species Profile:** The Grassland, Threats and Timeline & Status

Identify TWO distinct human impacts affecting the bandicoot’s grassland. For each: identify the type of impact, describe the mechanism of harm, and give specific evidence (a figure, a date or a documented case) of its effect on biodiversity or the bandicoot.

Impact	Type (e.g. habitat loss)	Mechanism of harm	Specific evidence
1			
2			

— Above The Level

Released: Lesson 10

Of the two impacts you identified, decide which is more serious for long-term biodiversity in your ecosystem. Justify your choice using evidence. Then propose ONE realistic policy change (at the local, state, or national level) that could reduce this impact, and explain the evidence that supports it.

11

Stage 11: Conservation & Action

Lesson concept: Strategies to protect biodiversity

Background

Conservation strategies aim to protect and restore biodiversity. **In-situ conservation** protects species in their natural habitat (e.g. national parks, Indigenous Protected Areas). **Ex-situ conservation** protects species outside their natural habitat (e.g. zoos, seed banks, captive breeding programs). The most effective conservation integrates both approaches alongside community action.

At The Level

Released: Lesson 11

Where to look: Species Profile: Timeline & Status

Part A — Find the conservation status of the Eastern Barred Bandicoot. Be careful: the *mainland* and *Tasmanian* populations are listed separately, with different statuses. Give the mainland figures.

Conservation status: _____

Population trend: _____ (increasing / decreasing / stable / unknown)

Where to look: Unit Booklet, Section 11 — Conservation & Action • Species Profile: Conservation and Timeline & Status

Part B — Choose ONE in-situ and ONE ex-situ conservation strategy in place for the bandicoot. Describe each and evaluate its effectiveness using specific evidence from the Species Profile.

Type	Strategy description	Evidence of effectiveness / limitations
In-situ		
Ex-situ		

— Above The Level

Released: Lesson 11

Where to look: Species Profile: Conservation, The Grassland and First Nations Knowledge

Design a realistic conservation action that students at your school or members of your local community could take to benefit the bandicoot or its ecosystem. Your response must address:

1. what the action is and how it would be carried out;
2. the ecological mechanism by which it helps the organism;
3. one limitation of your proposed action; and
4. how its success could be measured.

Where I am, and what would move me up

Criterion	My judgement	One thing that I could do better
Classification & naming		
Energy		
Food webs		
Ecosystems		
Biodiversity & invasive species		
Evaluating conservation (key criterion)		
Communicating science		

Teacher feedback